

Object Oriented Design Quality Metrics

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [jqassistant](#)
- [notebook walks through examples for integrating various packages with Neo4j](#)
- [OO Design Quality Metrics](#)
- [A Validation of Martin's Metric](#)
- [Neo4j Python Driver](#)

Incoming Dependencies

Incoming dependencies are also denoted as "Fan-in", "Afferent Coupling" or "in-degree". These are the ones that use the listed package.

If these packages get changed, the incoming dependencies might be affected by the change. The more incoming dependencies, the harder it gets to change the code without the need to adapt the dependent code ("rigid code"). Even worse, it might affect the behavior of the dependent code in an unwanted way ("fragile code").

Since Java Packages are organized hierarchically, incoming dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 1a

- Show the top 20 Java Packages with the most incoming dependencies
- Set the "incomingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incon
0	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	unitofwork	11695	41605	
1	axon-messaging-5.0.1	org.axonframework.messaging.core	core	10993	35618	
2	axon-common-5.0.1	org.axonframework.common.configuration	configuration	3336	17876	
3	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	3209	10719	
4	axon-common-5.0.1	org.axonframework.common.annotation	annotation	1842	2058	
5	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	1442	5110	
6	axon-common-5.0.1	org.axonframework.common.infra	infra	1212	1486	
7	axon-common-5.0.1	org.axonframework.common	common	838	2028	
8	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling	queryhandling	698	2468	
9	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	annotation	484	1624	
10	axon-conversion-5.0.1	org.axonframework.conversion	conversion	426	1719	
11	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	token	425	2089	
12	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	283	801	
13	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore	eventstore	223	719	
14	axon-messaging-5.0.1	org.axonframework.messaging.tracing	tracing	137	481	
15	axon-test-5.0.1	org.axonframework.test.fixture	fixture	132	357	
16	axon-modelling-5.0.1	org.axonframework.modelling	modelling	125	282	
17	axon-modelling-5.0.1	org.axonframework.modelling.entity	entity	115	394	
18	axon-modelling-5.0.1	org.axonframework.modelling.entity.child	child	89	230	
19	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	segmenting	86	659	

Table 1b

- Show the top 20 Java Packages including their sub-packages with the most incoming dependencies
- Set the property "incomingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incon
0	axon-messaging-5.0.1	org.axonframework.messaging.core	core	18122	61768	
1	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	unitofwork	11490	40479	
2	axon-common-5.0.1	org.axonframework.common	common	6759	22336	
3	axon-common-5.0.1	org.axonframework.common.configuration	configuration	3175	17216	
4	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	2081	7412	
5	axon-common-5.0.1	org.axonframework.common.annotation	annotation	1841	2057	
6	axon-common-5.0.1	org.axonframework.common.infra	infra	1198	1412	
7	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	883	3295	
8	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	annotation	396	1318	
9	axon-conversion-5.0.1	org.axonframework.conversion	conversion	374	1538	
10	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	token	350	1386	
11	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling	queryhandling	279	897	
12	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	231	656	
13	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	processing	217	663	
14	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	streaming	207	633	
15	axon-messaging-5.0.1	org.axonframework.messaging.tracing	tracing	108	359	
16	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore	eventstore	76	206	
17	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	segmenting	74	620	
18	axon-modelling-5.0.1	org.axonframework.modelling	modelling	74	226	
19	axon-modelling-5.0.1	org.axonframework.modelling.repository	repository	66	311	

Outgoing Dependencies

Outgoing dependencies are also denoted as "Fan-out", "Efferent Coupling" or "out-degree". These are the ones that are used by the listed package.

Code from other packages and libraries you're depending on (outgoing) might change over time. The more outgoing changes, the more likely and frequently code changes are needed. This involves time and effort which can be reduced by automation of tests and version updates. Automated tests are crucial to reveal updates, that change the behavior of the code unexpectedly ("fragile code"). As soon as more effort is required, keeping up becomes

difficult (“rigid code”). Not being able to use a newer version might not only restrict features, it can get problematic if there are security issues. This might force you to take “fast but ugly” solutions into account which further increases technical dept.

Since Java Packages are organized hierarchically, outgoing dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 2a

- Show the top 20 Java Packages with the most outgoing dependencies
- Set the "outgoingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgoin
0	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration	configuration	2549	15962	
1	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	pooled	2346	9920	
2	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore.jpa	jpa	1260	4805	
3	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.configuration	configuration	1237	4801	
4	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	event	1228	3848	
5	axon-messaging-5.0.1	org.axonframework.messaging.core.interception	interception	1084	5770	
6	axon-test-5.0.1	org.axonframework.test.fixture	fixture	954	3383	
7	axon-modelling-5.0.1	org.axonframework.modelling.entity.annotation	annotation	858	3265	
8	axon-modelling-5.0.1	org.axonframework.modelling.entity	entity	800	2887	
9	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	subscribing	737	2672	
10	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore	eventstore	705	2705	
11	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.autoconfig	autoconfig	678	2925	
12	axon-modelling-5.0.1	org.axonframework.modelling.configuration	configuration	585	2279	
13	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling	queryhandling	547	2129	
14	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.anno...	annotation	528	1611	
15	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.dist...	distributed	513	1412	
16	axon-messaging-5.0.1	org.axonframework.messaging.monitoring.interce...	interception	500	1550	
17	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.inte...	interception	491	1561	
18	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	annotation	464	1548	
19	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	438	1108	

Table 2b

- Show the top 20 Java Packages including their sub-packages with the most outgoing dependencies
- Set the property "outgoingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgo
0	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	656	2132	
1	axon-messaging-5.0.1	org.axonframework.messaging.core	core	398	1425	
2	axon-modelling-5.0.1	org.axonframework.modelling	modelling	381	1124	
3	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	processing	345	1285	
4	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing	eventsourcing	317	1060	
5	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	313	1019	
6	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling	queryhandling	310	1111	
7	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	streaming	275	1026	
8	axon-modelling-5.0.1	org.axonframework.modelling.entity	entity	270	856	
9	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	connector	253	870	
10	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	pooled	231	928	
11	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore	eventstore	183	597	
12	axon-modelling-5.0.1	org.axonframework.modelling.entity.annotation	annotation	179	681	
13	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	unitofwork	176	697	
14	axon-messaging-5.0.1	org.axonframework.messaging.core.interception	interception	166	587	
15	axon-test-5.0.1	org.axonframework.test	test	157	469	
16	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	event	142	420	
17	axon-test-5.0.1	org.axonframework.test.fixture	fixture	129	462	
18	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	annotation	124	337	
19	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conf...	configuration	120	382	

Instability

$$Instability = \frac{Outgoing\ Dependencies}{Outgoing\ Dependencies + Incoming\ Dependencies}$$

Instability is expressed as the ratio of the number of outgoing dependencies of a module (i.e., the number of packages that depend on it) to the total number of dependencies (i.e., the sum of incoming and outgoing dependencies).

Small values near zero indicate low *Instability*. With no outgoing but some incoming dependencies the *Instability* is zero which is denoted as maximally stable. Such code units are more rigid and difficult to change without impacting other parts of the system. If they are changed less because of that, they are considered stable.

Conversely, high values approaching one indicate high *Instability*. With some outgoing dependencies but no incoming ones the *Instability* is denoted as maximally unstable. Such code units are easier to change without affecting other modules, making them more flexible and less prone to cascading changes throughout the system. If they are changed more often because of that, they are considered unstable.

Since Java Packages are organized hierarchically, *Instability* can be calculated for every package in isolation or by including all of its sub-packages.

Table 3a

- Show the top 20 Java Packages with the lowest *Instability*
- Set the property "instability" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityF
0	axon-common-5.0.1	org.axonframework.common.annotation	annotation	0.001626	0.022556	0.000000	
1	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	unitofwork	0.010826	0.082397	0.153846	
2	axon-common-5.0.1	org.axonframework.common.infra	infra	0.021792	0.082645	0.100000	
3	axon-messaging-5.0.1	org.axonframework.messaging.core	core	0.030086	0.164609	0.178947	
4	axon-common-5.0.1	org.axonframework.common	common	0.035673	0.057627	0.000000	
5	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	token	0.036281	0.121212	0.200000	
6	axon-conversion-5.0.1	org.axonframework.conversion	conversion	0.046980	0.153846	0.250000	
7	axon-common-5.0.1	org.axonframework.common.configuration	configuration	0.074875	0.263959	0.295775	
8	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	0.120099	0.212435	0.392157	
9	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	0.192157	0.276119	0.444444	
10	axon-common-5.0.1	org.axonframework.common.util	util	0.200000	0.153846	0.000000	
11	axon-messaging-5.0.1	org.axonframework.messaging.core.conversion	conversion	0.247059	0.192308	0.571429	
12	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	store	0.263158	0.190476	1.000000	
13	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	0.270619	0.320513	0.461538	
14	axon-common-5.0.1	org.axonframework.common.jpa	jpa	0.272727	0.272727	1.000000	
15	axon-messaging-5.0.1	org.axonframework.messaging.monitoring	monitoring	0.282051	0.195122	0.666667	
16	axon-common-5.0.1	org.axonframework.common.jdbc	jdbc	0.310345	0.380952	0.000000	
17	axon-messaging-5.0.1	org.axonframework.messaging.tracing	tracing	0.331707	0.314815	0.533333	
18	axon-common-5.0.1	org.axonframework.common.lifecycle	lifecycle	0.333333	0.444444	0.000000	
19	axon-update-5.0.1	org.axonframework.update.api	api	0.379310	0.384615	0.000000	

Table 3b

- Show the top 20 Java Packages including their sub-packages with the lowest *Instability*
- Set the property "instabilityIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityF
0	axon-common-5.0.1	org.axonframework.common.annotation	annotation	0.001085	0.015267	0.000000	
1	axon-common-5.0.1	org.axonframework.common.infra	infra	0.004156	0.027523	0.000000	
2	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	unitofwork	0.015087	0.045643	0.500000	
3	axon-common-5.0.1	org.axonframework.common.configuration	configuration	0.021270	0.070796	0.425287	
4	axon-messaging-5.0.1	org.axonframework.messaging.core	core	0.021490	0.182213	0.389189	
5	axon-conversion-5.0.1	org.axonframework.conversion	conversion	0.057935	0.142857	0.500000	
6	axon-common-5.0.1	org.axonframework.common.util	util	0.090909	0.100000	0.000000	
7	axon-common-5.0.1	org.axonframework.common.jdbc	jdbc	0.133333	0.181818	0.000000	
8	axon-messaging-5.0.1	org.axonframework.messaging.core.conversion	conversion	0.136986	0.166667	0.500000	
9	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	token	0.152542	0.328358	0.500000	
10	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	0.163043	0.204082	0.500000	
11	axon-common-5.0.1	org.axonframework.common.lifecycle	lifecycle	0.166667	0.250000	0.000000	
12	axon-messaging-5.0.1	org.axonframework.messaging.tracing	tracing	0.205882	0.290323	0.461538	
13	axon-common-5.0.1	org.axonframework.common.jpa	jpa	0.222222	0.222222	0.000000	
14	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.230769	0.333333	0.500000	
15	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	annotation	0.238462	0.307692	0.421053	
16	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	0.239678	0.517413	0.573171	
17	axon-common-5.0.1	org.axonframework.common.property	property	0.250000	0.400000	0.000000	
18	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.util	util	0.250000	0.333333	0.000000	
19	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	0.261706	0.543478	0.527273	

Abstractness

$$Abstractness = \frac{\text{abstract classes in category}}{\text{total number of classes in category}}$$

Package *Abstractness* is expressed as the ratio of the number of abstract classes and interfaces to the total number of classes of a package.

Zero *Abstractness* means that there are no abstract types or interfaces in the package. On the other hand, a value of one means that there are only abstract types.

Since Java Packages are organized hierarchically, *Abstractness* can be calculated for every package in isolation or by including all of its sub-packages.

Table 4a

- Show the top 30 packages with the lowest *Abstractness*
- Set the property "abstractness" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
0	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.autoconfig	autoconfig	0.0	0	34
1	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot	springboot	0.0	0	21
2	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	event	0.0	0	14
3	axon-metrics-micrometer-5.0.1	org.axonframework.extension.metrics.micrometer	micrometer	0.0	0	12
4	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.inte...	interception	0.0	0	8
5	axon-messaging-5.0.1	org.axonframework.messaging.core.timeout	timeout	0.0	0	8
6	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.command	command	0.0	0	6
7	axon-messaging-5.0.1	org.axonframework.messaging.tracing.attributes	attributes	0.0	0	6
8	axon-update-5.0.1	org.axonframework.update.api	api	0.0	0	6
9	axon-tracing-opentelemetry-5.0.1	org.axonframework.extension.tracing.opentelemetry	opentelemetry	0.0	0	5
10	axon-common-5.0.1	org.axonframework.common.lifecycle	lifecycle	0.0	0	5
11	axon-messaging-5.0.1	org.axonframework.messaging.monitoring.interce...	interception	0.0	0	5
12	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.in...	interception	0.0	0	5
13	axon-conversion-5.0.1	org.axonframework.conversion.converter	converter	0.0	0	5
14	axon-conversion-5.0.1	org.axonframework.conversion.json	json	0.0	0	5
15	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	jpa	0.0	0	4
16	axon-update-5.0.1	org.axonframework.update.detection	detection	0.0	0	4
17	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.inte...	interception	0.0	0	3
18	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	inmemory	0.0	0	3
19	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration...	reflection	0.0	0	3
20	axon-test-5.0.1	org.axonframework.test.server	server	0.0	0	2
21	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.actuato...	axonserver	0.0	0	2
22	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	annotation	0.0	0	2
23	axon-messaging-5.0.1	org.axonframework.messaging.core.reflection	reflection	0.0	0	2
24	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.an...	annotation	0.0	0	2
25	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration	configuration	0.0	0	2
26	axon-common-5.0.1	org.axonframework.common.io	io	0.0	0	1
27	axon-common-5.0.1	org.axonframework.common.digest	digest	0.0	0	1
28	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.util.jpa	jpa	0.0	0	1
29	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.retry	retry	0.0	0	1

Table 4b

- Show the top 30 packages with the highest *Abstractness* and number of Java Types

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
141	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.actuator	actuator	1.000000	1	1
142	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	streaming	1.000000	1	1
139	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.co...	configuration	0.750000	3	4
140	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.conf...	configuration	0.750000	3	4
138	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conf...	configuration	0.714286	10	14
137	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.annotation	annotation	0.700000	7	10
135	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.util	util	0.571429	4	7
136	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.anno...	annotation	0.571429	4	7
134	axon-modelling-5.0.1	org.axonframework.modelling.repository	repository	0.555556	5	9
133	axon-common-5.0.1	org.axonframework.common.jdbc	jdbc	0.545455	6	11
132	axon-modelling-5.0.1	org.axonframework.modelling.configuration	configuration	0.538462	7	13
126	axon-common-5.0.1	org.axonframework.common.annotation	annotation	0.500000	2	4
127	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.tr...	transaction	0.500000	2	4
128	axon-common-5.0.1	org.axonframework.common.jpa	jpa	0.500000	1	2
129	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conv...	conversion	0.500000	1	2
130	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.gateway	gateway	0.500000	1	2
131	axon-messaging-5.0.1	org.axonframework.messaging.core.conversion	conversion	0.500000	1	2
125	axon-common-5.0.1	org.axonframework.common.configuration	configuration	0.478261	22	46
123	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	0.473684	9	19
124	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	0.473684	9	19
122	axon-messaging-5.0.1	org.axonframework.messaging.core.interception....	annotation	0.454545	5	11
121	axon-test-5.0.1	org.axonframework.test.fixture	fixture	0.451613	14	31
120	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	0.437500	7	16
118	axon-modelling-5.0.1	org.axonframework.modelling.entity.child	child	0.428571	6	14
119	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing	eventsourcing	0.428571	3	7
117	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling	queryhandling	0.416667	10	24
114	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.di...	distributed	0.400000	4	10
115	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.repl...	annotation	0.400000	4	10
116	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.gateway	gateway	0.400000	2	5
113	axon-messaging-5.0.1	org.axonframework.messaging.tracing	tracing	0.384615	5	13

Table 4c

- Show the top 30 packages including their sub-packages with the highest package depth and lowest *Abstractness*
- Set the property "abstractnessIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSub
0	axon-metrics-micrometer-5.0.1	org.axonframework.extension.metrics.micrometer	micrometer	0.0	0	13	
1	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration	configuration	0.0	0	5	
2	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.autoconfig	autoconfig	0.0	0	34	
3	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	event	0.0	0	14	
4	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.inte...	interception	0.0	0	8	
5	axon-messaging-5.0.1	org.axonframework.messaging.core.timeout	timeout	0.0	0	8	
6	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.command	command	0.0	0	6	
7	axon-messaging-5.0.1	org.axonframework.messaging.tracing.attributes	attributes	0.0	0	6	
8	axon-update-5.0.1	org.axonframework.update.api	api	0.0	0	6	
9	axon-tracing-opentelemetry-5.0.1	org.axonframework.extension.tracing.opentelemetry	opentelemetry	0.0	0	5	
10	axon-common-5.0.1	org.axonframework.common.lifecycle	lifecycle	0.0	0	5	
11	axon-messaging-5.0.1	org.axonframework.messaging.monitoring.interce...	interception	0.0	0	5	
12	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.in...	interception	0.0	0	5	
13	axon-conversion-5.0.1	org.axonframework.conversion.converter	converter	0.0	0	5	
14	axon-conversion-5.0.1	org.axonframework.conversion.json	json	0.0	0	5	
15	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	jpa	0.0	0	4	
16	axon-update-5.0.1	org.axonframework.update.detection	detection	0.0	0	4	
17	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.inte...	interception	0.0	0	3	
18	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	inmemory	0.0	0	3	
19	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration...	reflection	0.0	0	3	
20	axon-test-5.0.1	org.axonframework.test.server	server	0.0	0	2	
21	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.actuato...	axonserver	0.0	0	2	
22	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	annotation	0.0	0	2	
23	axon-messaging-5.0.1	org.axonframework.messaging.core.reflection	reflection	0.0	0	2	
24	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.an...	annotation	0.0	0	2	
25	axon-common-5.0.1	org.axonframework.common.io	io	0.0	0	1	
26	axon-common-5.0.1	org.axonframework.common.digest	digest	0.0	0	1	
27	axon-spring-boot-	org.axonframework.extension.springboot.util.jpa	jpa	0.0	0	1	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSub
	autoconfigure-5.0.1						
28	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.retry	retry	0.0	0	1	
29	axon-metrics-micrometer-5.0.1	org.axonframework.extension.metrics.micrometer...	reservoir	0.0	0	1	

Table 4d

- Show the top 30 packages including their sub-packages with the highest package depth and highest *Abstractness*

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	max5
112	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.co...	configuration	0.750000	3	4	
113	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.conf...	configuration	0.750000	3	4	
111	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conf...	configuration	0.714286	10	14	
110	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.anno...	annotation	0.571429	4	7	
109	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.annotation	annotation	0.562500	9	16	
108	axon-modelling-5.0.1	org.axonframework.modelling.repository	repository	0.555556	5	9	
107	axon-common-5.0.1	org.axonframework.common.jdbc	jdbc	0.545455	6	11	
106	axon-modelling-5.0.1	org.axonframework.modelling.configuration	configuration	0.538462	7	13	
99	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.util	util	0.500000	4	8	
100	axon-common-5.0.1	org.axonframework.common.annotation	annotation	0.500000	2	4	
101	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.tr...	transaction	0.500000	2	4	
102	axon-common-5.0.1	org.axonframework.common.jpa	jpa	0.500000	1	2	
103	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conv...	conversion	0.500000	1	2	
104	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.gateway	gateway	0.500000	1	2	
105	axon-messaging-5.0.1	org.axonframework.messaging.core.conversion	conversion	0.500000	1	2	
98	axon-common-5.0.1	org.axonframework.common.configuration	configuration	0.478261	22	46	
97	axon-messaging-5.0.1	org.axonframework.messaging.core.interception....	annotation	0.454545	5	11	
96	axon-test-5.0.1	org.axonframework.test.fixture	fixture	0.451613	14	31	
95	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	0.437500	7	16	
94	axon-modelling-5.0.1	org.axonframework.modelling.entity.child	child	0.428571	6	14	
91	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.di...	distributed	0.400000	4	10	
92	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.repl...	annotation	0.400000	4	10	
93	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.gateway	gateway	0.400000	2	5	
90	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling	queryhandling	0.380952	24	63	
89	axon-modelling-5.0.1	org.axonframework.modelling	modelling	0.376344	35	93	
87	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	0.375000	24	64	
88	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.ga...	gateway	0.375000	3	8	
86	axon-messaging-	org.axonframework.messaging.eventhandling.anno...	annotation	0.368421	7	19	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxS
	5.0.1						
84	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	unitofwork	0.363636	8	22	
85	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.dist...	distributed	0.363636	4	11	

Distance from the main sequence

The *main sequence* is a imaginary line that represents a good compromise between *Abstractness* and *Instability*. A high distance to this line may indicate problems. For example is very *stable* (rigid) code with low abstractness hard to change.

Read more details on that in [OO Design Quality Metrics](#) and [Calculate metrics](#).

Table 5a

- Show the top 30 packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-messaging-5.0.1	org.axonframework.messaging.tracing.attributes	attributes	NaN	0.0	NaN	6
1	axon-conversion-5.0.1	org.axonframework.conversion.converter	converter	NaN	0.0	NaN	5
2	axon-messaging-5.0.1	org.axonframework.messaging.core.reflection	reflection	NaN	0.0	NaN	2
3	axon-common-5.0.1	org.axonframework.common.io	io	NaN	0.0	NaN	1
4	axon-common-5.0.1	org.axonframework.common.digest	digest	NaN	0.0	NaN	1
5	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.actuator	actuator	NaN	1.0	NaN	1
6	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.retry	retry	NaN	0.0	NaN	1
7	axon-metrics-micrometer-5.0.1	org.axonframework.extension.metrics.micrometer...	reservoir	NaN	0.0	NaN	1
8	axon-tracing-opentelemetry-5.0.1	org	org	NaN	0.0	NaN	0
9	axon-tracing-opentelemetry-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
10	axon-tracing-opentelemetry-5.0.1	org.axonframework.extension	extension	NaN	0.0	NaN	0
11	axon-tracing-opentelemetry-5.0.1	org.axonframework.extension.tracing	tracing	NaN	0.0	NaN	0
12	axon-common-5.0.1	org	org	NaN	0.0	NaN	0
13	axon-common-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
14	axon-server-connector-5.0.1	org	org	NaN	0.0	NaN	0
15	axon-server-connector-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
16	axon-server-connector-5.0.1	org.axonframework.axonserver	axonserver	NaN	0.0	NaN	0
17	axon-modelling-5.0.1	org	org	NaN	0.0	NaN	0
18	axon-modelling-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
19	axon-eventsourcing-5.0.1	org	org	NaN	0.0	NaN	0
20	axon-eventsourcing-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
21	axon-test-5.0.1	org	org	NaN	0.0	NaN	0
22	axon-test-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
23	axon-spring-boot-autoconfigure-5.0.1	org	org	NaN	0.0	NaN	0
24	axon-spring-boot-autoconfigure-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
25	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension	extension	NaN	0.0	NaN	0
26	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.service	service	NaN	0.0	NaN	0
27	axon-messaging-5.0.1	org	org	NaN	0.0	NaN	0
28	axon-messaging-5.0.1	org.axonframework	axonframework	NaN	0.0	NaN	0
29	axon-messaging-5.0.1	org.axonframework.messaging	messaging	NaN	0.0	NaN	0

Table 5b

- Show the top 30 packages including their sub-packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-common-5.0.1	org.axonframework.common.lifecycle	lifecycle	0.833333	0.000000	0.166667	5
1	axon-conversion-5.0.1	org.axonframework.conversion	conversion	0.775399	0.166667	0.057935	30
2	axon-common-5.0.1	org.axonframework.common.infra	infra	0.745844	0.250000	0.004156	8
3	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	token	0.718425	0.129032	0.152542	31
4	axon-messaging-5.0.1	org.axonframework.messaging.core	core	0.694106	0.284404	0.021490	218
5	axon-update-5.0.1	org.axonframework.update.api	api	0.684211	0.000000	0.315789	6
6	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.conf...	configuration	0.671053	0.750000	0.921053	4
7	axon-test-5.0.1	org.axonframework.test.server	server	0.666667	0.000000	0.333333	2
8	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.co...	configuration	0.666667	0.750000	0.916667	4
9	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	unitofwork	0.621277	0.363636	0.015087	22
10	axon-common-5.0.1	org.axonframework.common.util	util	0.575758	0.333333	0.090909	6
11	axon-messaging-5.0.1	org.axonframework.messaging.tracing	tracing	0.530960	0.263158	0.205882	19
12	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.anno...	annotation	0.522648	0.571429	0.951220	7
13	axon-common-5.0.1	org.axonframework.common.configuration	configuration	0.500469	0.478261	0.021270	46
14	axon-update-5.0.1	org.axonframework.update.detection	detection	0.500000	0.000000	0.500000	4
15	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.util.jpa	jpa	0.500000	0.000000	0.500000	1
16	axon-update-5.0.1	org.axonframework.update.common	common	0.500000	0.000000	0.500000	1
17	axon-common-5.0.1	org.axonframework.common.annotation	annotation	0.498915	0.500000	0.001085	4
18	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.annotation	annotation	0.473214	0.562500	0.910714	16
19	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	eventhandling	0.444532	0.315789	0.239678	171
20	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	annotation	0.441538	0.320000	0.238462	50
21	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.435897	0.333333	0.230769	3
22	axon-common-5.0.1	org.axonframework.common.property	property	0.416667	0.333333	0.250000	9
23	axon-modelling-5.0.1	org.axonframework.modelling.configuration	configuration	0.415174	0.538462	0.876712	13
24	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.actuato...	axonserver	0.400000	0.000000	0.600000	2
25	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.gateway	gateway	0.400000	0.500000	0.900000	2
26	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	0.399457	0.437500	0.163043	16
27	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conf...	configuration	0.377269	0.714286	0.662983	14
28	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling	commandhandling	0.363294	0.375000	0.261706	64
29	axon-messaging-5.0.1	org.axonframework.messaging.core.conversion	conversion	0.363014	0.500000	0.136986	2

Abstractness vs. Instability Plot with "Main Sequence" line as reference

- Plot *Abstractness* vs. *Instability* of all packages

- Draw the "main sequence" as dashed green diagonal line
- Scale the packages by the number of types they contain
- Color the packages by their distance to the "main sequence" (blue=near, red=far)

Figure 5a - Packages without their sub-packages

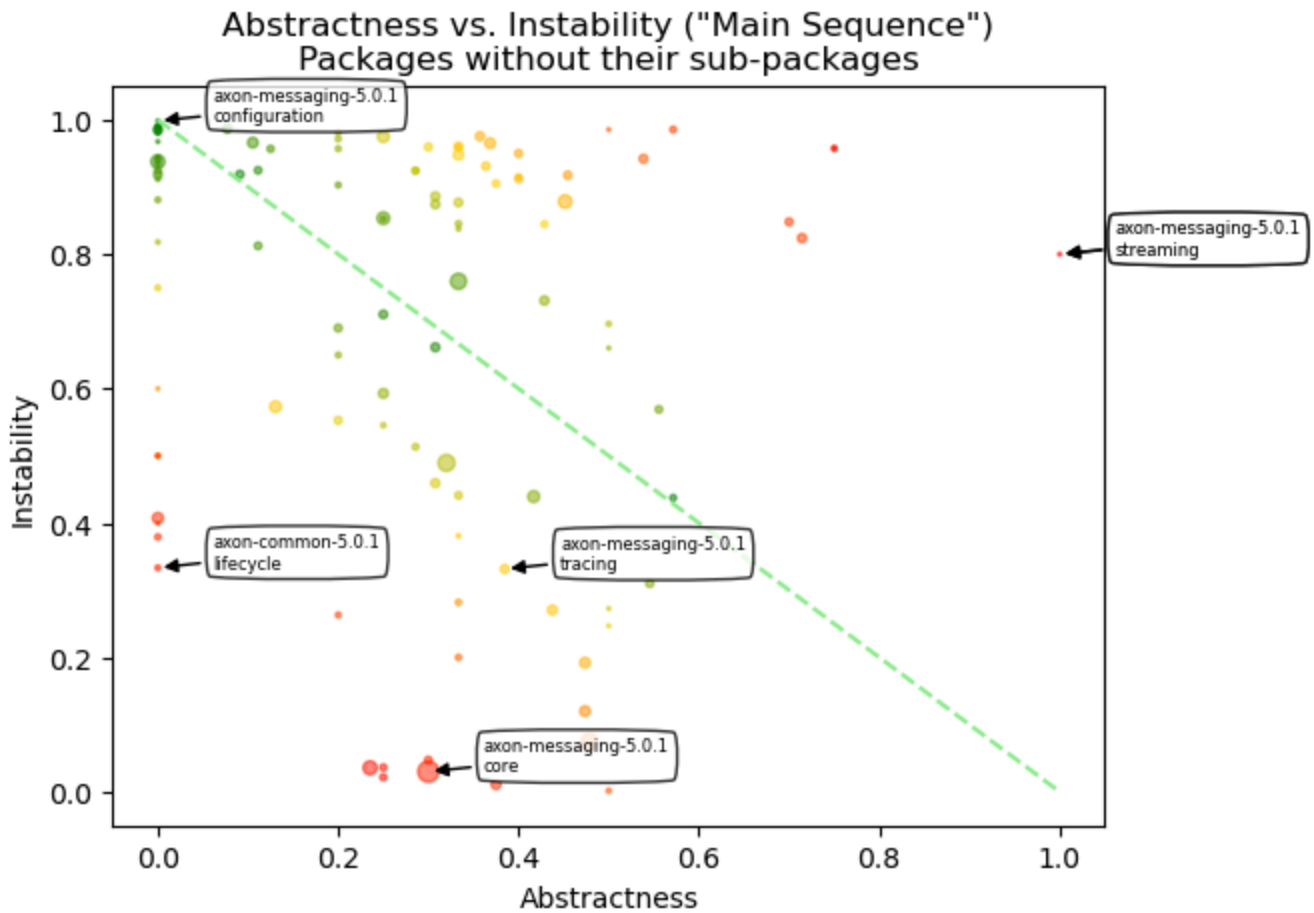


Figure 5b - Packages including their sub-packages

Abstractness vs. Instability ("Main Sequence")
Packages including their sub-packages

